EXAM 2

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Question 1

1 pt 1

For a standard normal distribution, find:

$P(z < c) = 0.9001$

Find c rounded to two decimal places.

Question 2

1 pt 1

For a confidence level of 90%, find the critical value

Question 3

5 pts 1

The numbers of false fire alarms were counted each month at a number of sites. The results are given in the following table.

Month	Number of Alarms
January	44
February	43
March	33
April	41
May	35
June	26
July	37
August	38
September	26
October	29
November	44
December	43

Test the hypothesis that false alarms are equally likely to occur in any month. Use 5% level of significance.

Procedure: Select an answer

Assumptions: (select everything that applies)

- ☐ At most 20% of the expected frequencies are less than 5
- ☐ Independent samples
- ☐ Equal population standard deviations
- ☐ Normal populations
- ☐ All expected frequencies are 1 or greater
- ☐ Simple random sample

Question 4 1 pt 1

According to the CDC, the average weight of men in 1990 was 181 pounds.

Consider a claim that the average weight of men increased by 2014.

(https://abcnews.go.com/Health/average-weight-american-men-15-pounds-20-years/story?id=41100782)

Use these hypotheses:

H0: mean = 181

HA: mean > 181

If you fail to reject the null hypothesis, then your conclusion would be:

- There is sufficient evidence to warrant rejection of the null hypothesis and to support the claim that the average weight of men increased.
- There is not sufficient evidence to warrant rejection of the null hypothesis that the average weight of men is the same.

Question 5

1 pt 1

You are conducting a study to see if the average male life expectancy is significantly different from 96. If your null and alternative hypothesis are:

$H_0: \mu = 96$
$H_1: \mu \neq 96$

Then the test is:

- right tailed
- left tailed
- two tailed

Question 6

1 pt 1

A statistics instructor believes that fewer than 20% of Evergreen Valley College (EVC) students attended the opening night midnight showing of the latest Harry Potter movie. She surveys 84 of her students and finds that 11 attended the midnight showing. An appropriate alternative hypothesis is:

- p = 0.20
- p ≤ 0.20
- p < 0.20
- p > 0.20

Question 7 1 pt 1

You are conducting a study to see if the typical doctor's salary (in thousands of dollars) is significantly less than 92. If your null and alternative hypothesis are:

$H_0: \mu = 92$
$H_1: \mu < 92$

Then the test is:

- right tailed
- two tailed
- left tailed

Question 8 1 pt 1

You are conducting a study to see if the probability of a true negative on a test for a certain cancer is significantly different from 0.39. You use a significance level of `alpha = 0.10`.

`H_0 : p = 0.39`
`H_1 : p != 0.39`

You obtain a sample of size `n=685` in which there are 245 successes.

What is the test statistic for this sample? (Report answer accurate to three decimal places.)

test statistic = ______

What is the p-value for this sample? (Report answer accurate to four decimal places.)

p-value = ______

The p-value is...

- less than (or equal to) `alpha`
- greater than `alpha`

This test statistic leads to a decision to...

- reject the null
- accept the null
- fail to reject the null

As such, the final conclusion is that...

- There is sufficient evidence to warrant rejection of the claim that the probability of a true negative on a test for a certain cancer is different from 0.39.
- There is not sufficient evidence to warrant rejection of the claim that the probability of a true negative on a test for a certain cancer is different from 0.39.
- The sample data support the claim that the probability of a true negative on a test for a certain cancer is different from 0.39.
- There is not sufficient sample evidence to support the claim that the probability of a true negative on a test for a certain cancer is different from 0.39.

Question 9

1 pt 1

If the probability (p-value) is greater than the level of significance, what decision is made in the hypothesis test?

- Accept the null hypothesis as true
- Reject the null hypothesis
- Fail to reject the null hypothesis

Question 10 1 pt 1

In a hypothesis test, a researcher obtains a p-value= 0.5814. The level of significance= 0.001. Give the correct conclusion.

- Fail to reject the null hypothesis
- Accept the alternative hypothesis
- Reject the null hypothesis

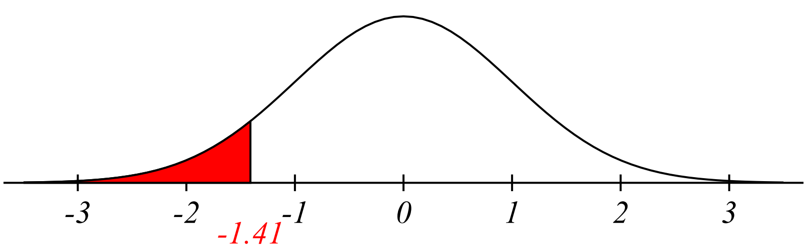
Question 11 1 pt 1

For a standard normal distribution, find:

P(-2.23 < z < -0.75)

Question 12 1 pt 1

The critical region and critical value for a hypothesis test are shown below. The test statistic is Z = -1.7. Should you reject or fail to reject the null hypothesis.



- Reject
- Fail to Reject

Question 13 1 pt 1

What specifically is used to decide whether to reject or fail to reject the null hypothesis?

- To reject the null hypothesis we need a better alternative hypothesis.
- To reject the null hypothesis we need a sufficiently small p-value.
- To reject the null hypothesis we need a sufficiently large p-value.
- To reject the null hypothesis we need a sufficiently small significance level.

Question 14 1 pt 1

What is the p-value of a hypothesis test?

- A p-value is the probability of finding another random sample with a statistic that is more extreme than the current sample statistic, given that the null hypotheses is wrong.
- A p-value is the probability of finding another random sample with a statistic that is more extreme than the current sample statistic, given that the null hypothesis is correct.
- A p-value is the probability that your sample statistic is less than the population parameter.
- A p-value is a probability that is smaller than the significance level.

Question 15

1 pt 1

How do you tell whether the test is left, right, or two tailed?

- It depends on the alternative hypothesis.
  Left tail for `<`
  Right tail for `>`
  Two tail for `!=`
- It depends on the null hypothesis.
  Left tail for `<=`
  Right tail for `>=`
  Two tail for `=`

Question 16

1 pt 1

You are conducting a study to see if the mean weight of goats (in pounds) is significantly less than 89. Thus you are performing a left-tailed test. The test significance level is `alpha` = .1.

The test statistic, t, has a p-value = 0.14.

Should you reject or fail to reject the null hypothesis? 1

- reject
- fail to reject

Question 17

3 pts 1

The US Department of Energy reported that more than 61% of homes were heated by natural gas. A random sample of 211 homes in Oregon found that 142 were heated by natural gas. Test the claim using a 1% significance level.

a. What type of test will be used in this problem?
Select an answer

b. What evidence justifies the use of this test? Check all that apply

- The sample size is larger than 30
- The original population is approximately normal
- The population standard deviation is not known
- There are two different samples being compared
- The population standard deviation is known
- `np > 5` and `nq > 5`
- The sample standard deviation is not known

c. Enter the null hypothesis for this test.
`H_0` ? ? __________

d. Enter the alternative hypothesis for this test.
`H_1:` ? ? __________

e. Is the original claim located in the null or alternative hypothesis?
Select an answer

f. What is the test statistic for the given statistics?

____

g. What is the *p*-value for this test?

____

h. What is the decision based on the given statistics?
Select an answer

i. What is the correct interpretation of this decision?

Using a _____ % level of significance, there Select an answer sufficient evidence to Select an answer the claim that more than 61% of homes were heated by natural gas.

Question 18 5 pts 1

An incubation period is a time between when you contract a virus and when your symptoms start. By surveying randomly selected local hospitals, a researcher was able to obtain the sample of incubation periods of 31 patients. The sample of incubation periods has a mean of 6.06 days and a standard deviation of 1.84 days. Construct a 90% confidence interval for the average incubation period of the novel coronavirus.

i. Procedure: Select an answer

ii. Assumptions: (select everything that applies)

- ☐ Simple random sample
- ☐ Sample size is greater than 30
- ☐ The number of positive and negative responses are both greater than 10
- ☐ Population standard deviation is known
- ☐ Population standard deviation is unknown
- ☐ Normal population

iii. Unknown parameter: Select an answer

Question 19 1 pt 1

You are conducting a test of homogeneity for the claim that two different populations have the same proportions of the following two characteristics. Here is the sample data.

Category	Population #1	Population #2
A	20	41
B	35	34

The expected observations for this table would be

Category	Population #1	Population #2
A		
B		

The resulting squared Pearson residuals are:

Category	Population #1	Population #2
A		
B		

What is the chi-square test-statistic for this data?

\`chi^2 = \`

Report all answers accurate to three decimal places. (Though you may need to retain more decimal places in the residuals to get the correct chi-square test statistic.)

Hint 1

Question 20 1 pt 1

A soft drink filling machine, when working properly, fills bottles with an average of 12 ounces of soft drink.

If the machine is either over-filling or under-filling, the machine should be shut down and readjusted. You are a quality control employee responsible for monitoring this particular machine and test if the machine works properly.

a. $H_0: \mu = 12$.
This null hypothesis implies
Select an answer

b. Select the correct sign for the alternative hypothesis and its meaning. $H_a: \mu$? 12.
This alternative hypothesis implies
Select an answer

c. In this context describe a **Type I error** and the **impact such an error** would have on the company.

- When the machine does not work properly it is not shut down and drinks not filled properly are manufactured. The customers purchased underfilled or overfilled drinks.
- When the machine works properly it is shut down and readjusted. This is an unnecessary loss to the company.

d. In this context describe a **Type II error** and the **impact such an error** would have on the company.

- When the machine does not work properly it is not shut down and drinks not filled properly are manufactured. The customers purchased underfilled or overfilled drinks.
- When the machine works properly it is shut down and readjusted. This is an unnecessary loss to the company.

e. The **significance level** is probability of making which type of error?

- Type II Error
- Type I Error

Question 21 5 pts 1

Part 1 of 5

For a certain drug, based on standards set by the United States Pharmacopeia (USP) - an official public standards-setting authority for all prescription and over-the-counter medicines and other health care products manufactured or sold in the United States, a standard deviation of capsule weights of less than 2 mg is acceptable. A sample of 15 capsules was taken and the weights are provided below:

100.8	98.6	99.4	100.5	100.1
100.1	100.7	101.2	99.9	99
99.6	100.1	99.5	99.4	99.9

The mean of the data set is 99.92; the sample standard deviation is 0.7; if the normality plot is not provided you may assume that the capsule weights are normally distributed.

Construct a 99% confidence interval for the variance (standard deviation) of all capsule weights.

i. Procedure: Select an answer

ii. Assumptions: (select everything that applies)

- The number of positive and negative responses are both greater than 10
- Simple random sample
- Population standard deviation is known
- Population standard deviation is unknown
- Sample size is greater than 30
- Normal population

iii. Unknown parameter: Select an answer

Question 22

1 pt 1

State the indicated conclusion in nontechnical terms, being sure to address the original claim.

An entomologist writes an article in a scientific journal claiming that **at least** 1.22% of male fireflies are unable to produce light due to a genetic mutation.

A hypothesis test of the claim has been conducted, and the decision from that test is to **reject** H_0.

- There **is** sufficient evidence to **support** the claim that the true proportion is at least 1.22%.
- There is **not** sufficient evidence to warrant **rejection** of the claim that the true proportion is at least 1.22%.
- There **is** sufficient evidence to warrant **rejection** of the claim that the true proportion is at least 1.22%.
- There is **not** sufficient evidence to **support** the claim that the true proportion is at least 1.22%.